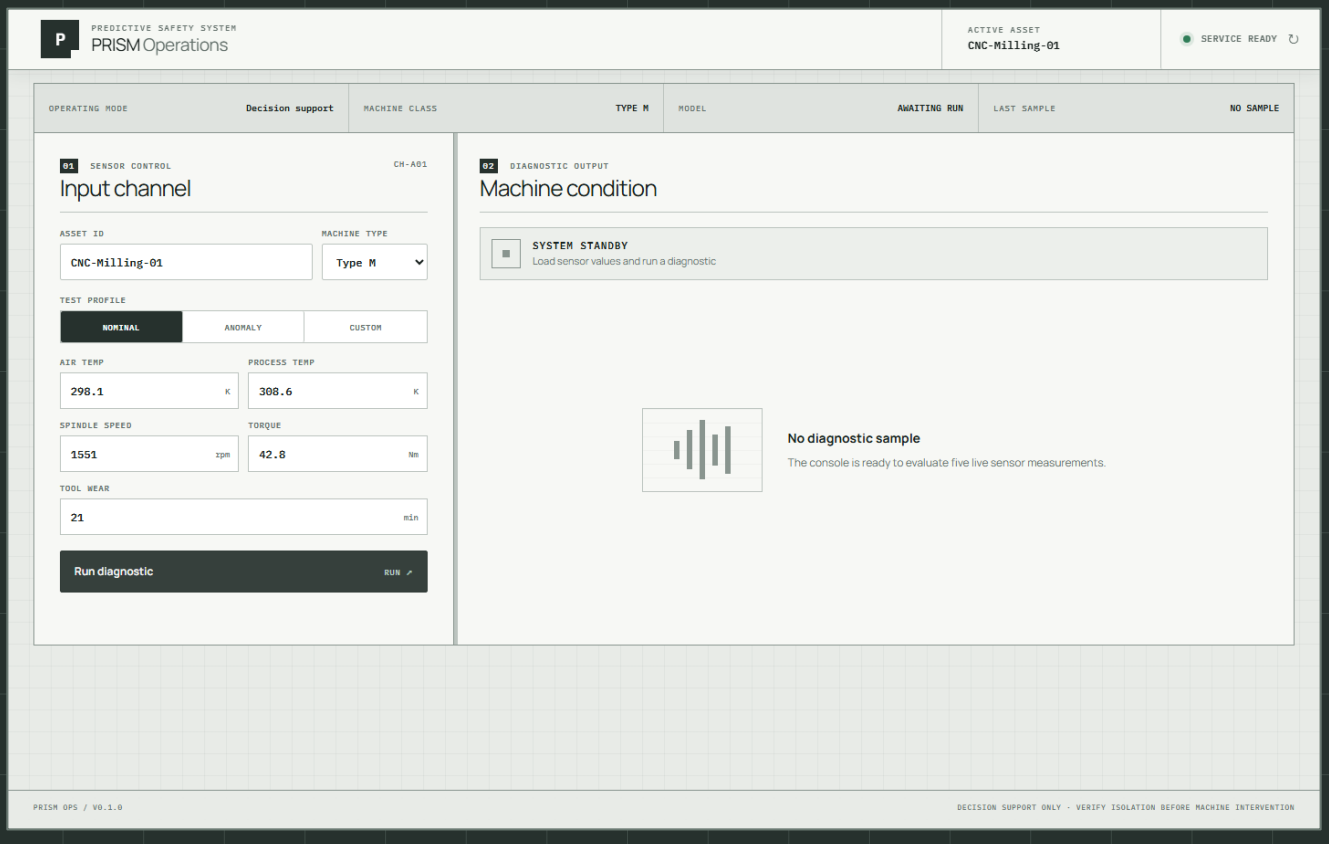
P.R.I.S.M.

Predictive Retrieval for Industrial Safety & Maintenance

An offline-first system that turns machine signals into traceable, validated safety guidance.



PREDICTIVE ML | HYBRID RAG | LOCAL LLM | INDUSTRIAL HMI

MIFTAH RIZKY

ML & FULL-STACK ENGINEERING

01 / THE PROBLEM

An alarm is not an action plan.

Industrial telemetry can reveal that something is wrong—yet still leave the operator without context, evidence, or a safe next step.

01	SIGNAL	**Sensor values**	Measurements arrive continuously, but raw numbers do not explain failure risk.
02	DECISION	**Failure context**	A useful system must distinguish nominal operation from likely failure signatures.
03	ACTION	**Verified guidance**	Safety recommendations need source evidence, validation, and human authorization.

SAFETY BOUNDARY

Decision support only. PRISM never commands machinery, PLCs, or safety systems.

Project scope and safety constraints: README.md

02 / WHAT PRISM DOES

One request. Two intentional paths.

A calibrated probability gate keeps nominal responses fast and reserves retrieval plus generation for critical conditions.

FIVE MACHINE MEASUREMENTS

AIR TEMP | PROCESS TEMP | SPEED | TORQUE | TOOL WEAR

Predictive engine

Calibrated failure probability + multi-label failure signatures

BELOW THRESHOLD

HEALTHY

Return ML result immediately.
Retrieval and generation are skipped.

AT OR ABOVE THRESHOLD

CRITICAL

Classify → retrieve evidence → generate → validate before display.

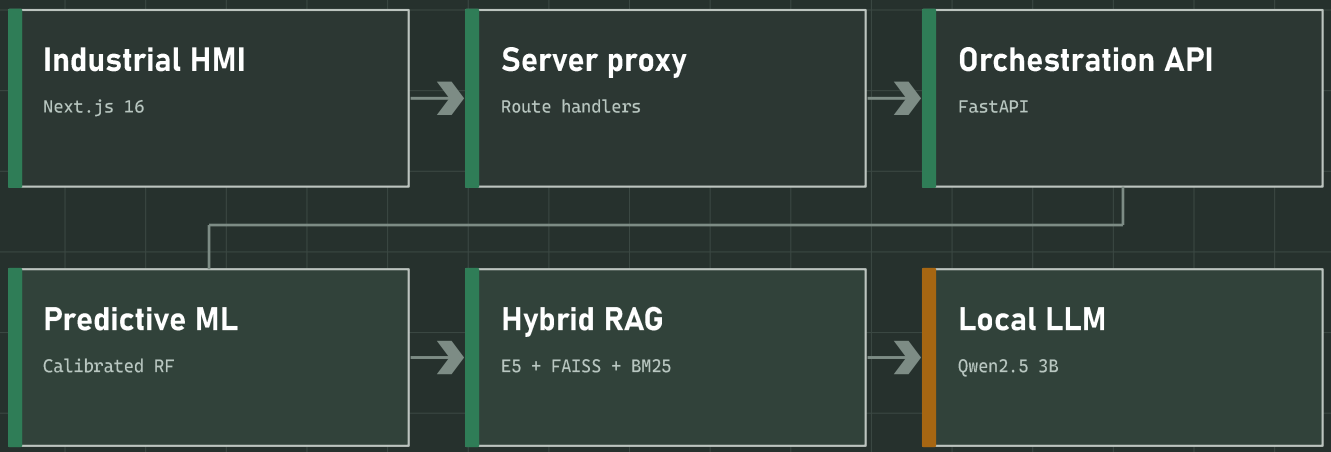
OPERATIONAL THRESHOLD `0.01326`

Model threshold and routing: artifacts/model_metadata.json, src/prism/api/

03 / ARCHITECTURE

Local-first by design.

The browser reaches a single orchestration boundary. Models, evidence, generation, and validation remain inside the local runtime.



MODEL ARTIFACTS

Versioned bundle, metadata, thresholds

VERIFIED SOURCES

OSHA PDFs, page metadata, checksums

`VALIDATION GATE → MITIGATION_READY / MANUAL_REVIEW_REQUIRED`

Runtime architecture: README.md and compose.yaml

04 / EVIDENCE

Measured, not imagined.

The portfolio claims below are taken directly from committed model, retrieval, and acceptance artifacts.

0.9604

HOLDOUT ROC-AUC

2,000-row holdout set

91.18%

FAILURE RECALL

62 of 68 failures detected

100%

RAG RECALL@5

8/8 evaluation queries hit

100%

SCHEMA VALIDATION

Deterministic acceptance run

715.238 ms

ANOMALY ORCHESTRATION P95

100 deterministic critical-path runs; target < 3 seconds

BENCHMARK SCOPE

This measures orchestration with a deterministic valid generator—not live Qwen/Ollama inference.

Sources: artifacts/acceptance_report.json and artifacts/model_metadata.json

05 / GUARDRAILS

Generation is the last step— not the source of truth.

Critical guidance must remain traceable to retrieved safety evidence and survive strict machine validation.

01	RETRIEVE	Multilingual E5 FAISS IndexFlatIP
02	FUSE	BM25 + dense results Reciprocal Rank Fusion
03	GENERATE	Qwen2.5 3B Local through Ollama
04	VALIDATE	Pydantic schema Citation grounding

334 PAGE-PRESERVING CHUNKS

4 VERIFIED OSHA PUBLICATIONS

LOCKOUT STATE

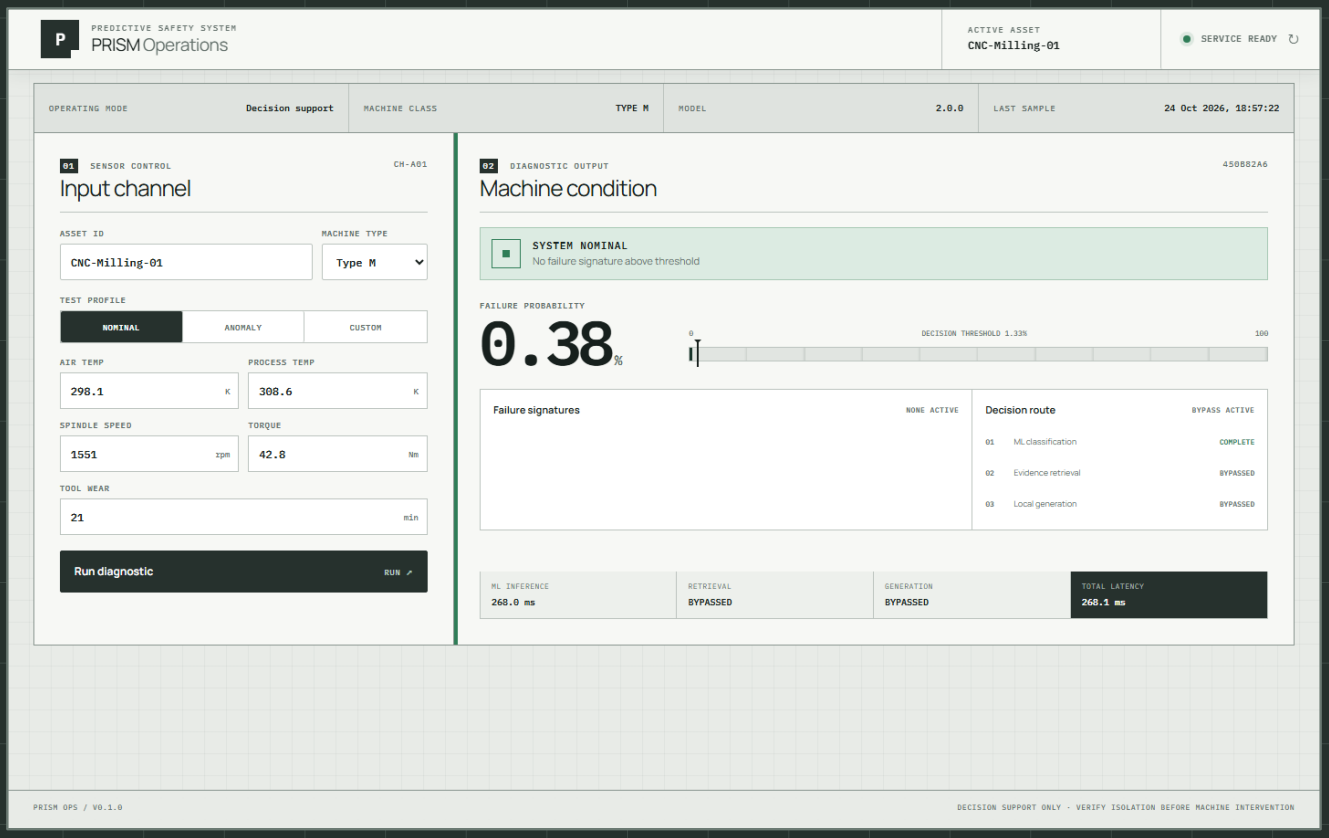
MANUAL REVIEW

Sources: knowledge index metadata and RAG evaluation report

06 / INDUSTRIAL HMI

Designed for the first operational glance.

The interface exposes readiness, sensor bounds, risk, routing, and latency without inventing fleet or historical data.



API READY

Connection and model readiness

SYSTEM NOMINAL

Probability below learned threshold

STAGE TELEMETRY

ML, retrieval, generation, total

REAL LOCAL DASHBOARD / 1440 × 900 CAPTURE

Source image: docs/assets/prism-dashboard-nominal.png

07 / DELIVERY

Built as a system, not a notebook demo.

Training, inference, knowledge retrieval, validation, deployment, and UI are versioned as one reproducible project.

MODEL

Deterministic splits
Candidate comparison
Calibrated probabilities

QUALITY

Unit + integration
Acceptance + fault injection
Frontend production build

OPERATIONS

Docker Compose
Persistent model caches
NVIDIA runtime profile

DELIVERY

GitHub Actions
Non-root containers
Operations runbook

TECH STACK

Python · scikit-learn · XGBoost · FAISS · FastAPI · Ollama

Next.js 16 · React 19 · TypeScript 5 · Docker Compose

CI GATE → PYTHON LINT + TESTS + ACCEPTANCE | WEB LINT + BUILD

Sources: pyproject.toml, web/package.json, compose.yaml, .github/workflows/

P.R.I.S.M.

Prediction with context.
Evidence before guidance.
Human authorization by design.

PREDICT	Calibrated machine-failure risk
TRACE	Verified page-level safety evidence
VALIDATE	Schema and citation guardrails
OPERATE	Purpose-built industrial HMI

EXPLORE THE PROJECT

Architecture, benchmarks, source code.

github.com/mifthrizky/P.R.I.S.M.

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ML & FULL-STACK ENGINEERING